

Kenji Tojo

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About

Kenji Tojo received his Ph.D. from the University of Tokyo, specializing in computer graphics and computational design. He has published multiple first-author papers at top venues, including SIGGRAPH, and his research has been recognized with several awards, such as the Dean's Prize for Outstanding Master's and Doctoral Theses and a Best Paper Honorable Mention at SGP. His work bridges computer graphics, computer vision, and machine learning, with a particular focus on extending inverse rendering and differentiable graphics techniques to support a broader range of applications, including content creation and design optimization.

Education

Ph.D. Information Science and Technology – The University of Tokyo	<i>Apr 2023 – Mar 2026</i>
- Adviser: Nobuyuki Umetani - Dean's prize for best Ph.D. thesis	
M.S. Information Science and Technology – The University of Tokyo	<i>Apr 2021 – Mar 2023</i>
- Adviser: Nobuyuki Umetani - Dean's prize for best M.S. thesis	
B.S. Information Science – The University of Tokyo	<i>Apr 2017 – Mar 2021</i>
Thesis Adviser: Takeo Igarashi	

Publications

DiffSoup: Direct Differentiable Rasterization of Triangle Soup for Extreme Radiance Field Simplification	<i>Jun 2026</i>
Kenji Tojo, Bernd Bickel, Nobuyuki Umetani <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026</i>	
Strands2Cards: Automatic Generation of Hair Cards from Strands	<i>Dec 2025</i>
Kenji Tojo, Liwen Hu, Nobuyuki Umetani, Hao Li <i>SIGGRAPH Asia 2025 Conference Proceedings</i>	
The Mokume Dataset and Inverse Texturing of Solid Wood	<i>Aug 2025</i>
- Maria Larsson, Hodaka Yamaguchi, Ehsan Pajouheshgar, I-Chao Shen, Kenji Tojo, Chia-Ming Chang, Lars Hansson, Olof Broman, Takashi Ijiri, Ariel Shamir, Wenzel Jakob, Takeo Igarashi <i>ACM Transactions on Graphics (presented at SIGGRAPH 2025)</i>	
GreenCloud: Volumetric Gradient Filtering via Regularized Green's Functions	<i>Jul 2025</i>
Kenji Tojo, Nobuyuki Umetani <i>Symposium of Geometry Processing (SGP) 2025 – Best Paper Honorable Mention</i>	
3D Gabor Splatting: Reconstruction of High-frequency Surface Texture using Gabor Noise	<i>Apr 2025</i>
- Haato Watanabe, Kenji Tojo, Nobuyuki Umetani <i>Eurographics 2025 – Short Papers</i>	
Free-form Floor Plan Design using Differentiable Voronoi Diagram	<i>Oct 2024</i>
- Xuanyu Wu, Kenji Tojo, Nobuyuki Umetani <i>Pacific Graphics 2024</i>	
Fabricable 3D Wire Art	<i>Aug 2024</i>
Kenji Tojo, Ariel Shamir, Bernd Bickel, Nobuyuki Umetani <i>SIGGRAPH 2024 Conference Proceedings</i>	

Stealth Shaper: Reflectivity Optimization as Surface Stylization	<i>Aug 2023</i>
◦ Kenji Tojo, Ariel Shamir, Bernd Bickel, Nobuyuki Umetani	
◦ SIGGRAPH 2023 Conference Proceedings	
Recolorable Posterization of Volumetric Radiance Fields Using Visibility-Weighted Palette Extraction	<i>Jul 2022</i>
◦ Kenji Tojo, Nobuyuki Umetani	
◦ Eurographics Symposium on Rendering (EGSR) 2022	
Neural Motion Compression with Frequency-adaptive Fourier Feature Network	<i>Apr 2022</i>
◦ Kenji Tojo, Nobuyuki Umetani	
◦ Eurographics 2022 – Short Papers	

Awards

Dean's Prize for Outstanding Doctoral Research – The University of Tokyo	<i>Mar 2026</i>
◦ Awarded for the best Ph.D. thesis of the year in the Creative Informatics Department.	
Best Paper Honorable Mention – SGP 2025	<i>Jul 2025</i>
◦ Recognized for the paper <i>GreenCloud: Volumetric Gradient Filtering via Regularized Green's Functions</i> .	
JSPS Fellowship Supplement for Outstanding Research Progress	<i>Apr 2025 – Mar 2026</i>
◦ Awarded a competitive annual supplement (360,000 JPY) in recognition of exceptional research performance during JSPS Fellowship.	
Japan Society for the Promotion of Science (JSPS) Research Fellowship	<i>Apr 2023 – Mar 2026</i>
◦ Awarded a prestigious national fellowship providing 2,400,000 JPY annually to support doctoral research.	
Dean's Prize for Outstanding Master's Research – The University of Tokyo	<i>Mar 2023</i>
◦ Awarded for the best Master's thesis of the year in the Creative Informatics Department.	

Experience

Visiting Researcher – ETH Zürich, Zürich, Switzerland	<i>Jun 2025 – Aug 2025</i>
◦ Mentor: Bernd Bickel	
Visiting Researcher – MBZUAI, Abu Dhabi, United Arab Emirates	<i>Sep 2024 – Nov 2024</i>
◦ Mentor: Hao Li	
Visiting Researcher – ISTA, Klosterneuburg, Austria	<i>Dec 2023 – Jan 2024</i>
◦ Mentor: Bernd Bickel	
Software Engineer Intern – Morgenrot Inc., Tokyo, Japan	<i>Mar 2023 – Mar 2023</i>
Research Assistant – The University of Tokyo, Tokyo, Japan	<i>Feb 2021 – Mar 2021</i>
◦ Mentor: Takeo Igarashi	

Teaching

CV 802: ADVANCED 3D COMPUTER VISION – MBZUAI	<i>Autumn, 2024</i>
◦ Invited Lecturer – Topic: <i>Differentiable Rendering</i>	

Technical skills

Programming: C++, OpenGL, Eigen, CUDA, Python, PyTorch, pybind11, nanobind etc.
Creative: Adobe Illustrator, Adobe Premiere Pro, Blender, Unreal Engine, etc.

Test scores

TOEFL iBT: 105 (October 15, 2022)